

The Renaissance of the Substrate: why the vacuum is not empty

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January 1, 2026

Standalone article extending the treatise *Time May Not Exist!* (Brécheteau, 2025)

DOI: 10.5281/zenodo.17214502

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Abstract

This article rehabilitates the notion of a physical background through the concept of the Chronon Field $\Phi(x)$, introduced in the treatise *Time May Not Exist!*. Against the idea of a vacuum reduced to geometry, it proposes to reread the substrate as the rhythmic fabric of the real: a non-energetic yet vibratory background, where the coherence of the world stems from a fundamental pulsation. By confronting approaches from relativity, quantum mechanics, and vacuum theory, it shows that “nothingness” is in fact an active field of local synchronization. The Chronon Field thus appears as a condition for the holding of the real, offering a new path between structure and becoming, between law and rhythm. At the operational level, we specify the dimension $[\Phi] = \text{s}^{-1}$ and the coupling hypothesis $\Gamma(x) \propto \Phi(x)$ linking decoherence to the local tempo, opening the way to metrological tests via clocks, spectroscopy, and interferometry.

Keywords: Chronon Field, rhythmic substrate, quantum vacuum, ether, coherence, background physics, philosophy of time

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I — The forgetting of the background

For a century, physics has chased away the word “ether” as superstition. It has been treated as a Newtonian ghost, a leftover from a world where light would need a medium in which to travel. With Einstein, the vacuum became noble: propagation no longer occurred in a substance, but in a geometry. The electromagnetic field, spacetime, and then quanta replaced support with structure. The world seemed to stand without a background, suspended in pure relation.

But this victory of form over matter has a cost. By removing the support, we lost the *holding*. Science gained in mathematical elegance what it lost in physical intuition. The vacuum became conceptually clean — but existentially hollow. The equations stand, yet our understanding of the real wavers, as if we had confused formal coherence with ontological stability.

Today, this forgetting comes back to haunt theory. Because the quantum vacuum is not empty: it murmurs, it fluctuates, it *beats*. It is no longer an absence, but a field of permanent activity, whose detectors and fluctuation effects betray its presence. It is here that the *Chronon Field* proposes a decisive shift: the return of the *background*, not as matter, but as *rhythm*.

Terminology. — Throughout this text, the word *substrate* designates neither a mechanical fluid nor a background energy; it refers to a *condition of cadence and coherence* (a *non-energetic* and *non-metric* substrate) carried by $\Phi(x)$, so as to avoid any ambiguity with a modification of the metric or the addition of a $T_{\mu\nu}$.

II — The full vacuum

Physicists know it well: what we call “vacuum” is a saturated theatre. Quantum fields are never silent: they fluctuate, exchange, and imperceptibly shift the energy levels and effective pressures measured in the

laboratory. The Lamb shift (1947) attests that the hydrogen atom is not insensitive to vacuum fluctuations; the Casimir effect reveals a pressure between closely spaced conducting plates; attophysics probes ultra-fast dynamics down to the zeptosecond scale. None of this postulates a “substance”: these are *metrological signatures* of an underlying agitation.

In the standard reading, these fluctuations are deemed “background noise”: a statistical agitation without further status. We propose a minimal and testable shift: to read in them the *signature of a rhythm of holding*. Not an additional energy, nor an ether fluid, but a *local coherence cadence* that sets the possibility of appearing and persisting.

This is precisely the role assigned to the *Chronon Field* $\Phi(x)$: a *rhythmic, non-energetic, non-metric substrate* (dimension $[\Phi] = \text{s}^{-1}$) that reparametrizes *cadences* without altering geometry or adding any $T_{\mu\nu}$. The vacuum thus becomes *full of coherence*, not of its own energy: it “holds” by a measurable pulsation. Just as a reference oscillator makes the atomic clock possible, $\Phi(x)$ locally ensures the synchronization that makes propagation, conservation, and the stability of standards possible.

III — The death and resurrection of the ether

The classical ether was sacrificed for good reasons: it postulated a preferred frame, contrary to relativity. But to confuse the abolition of a *mechanical medium* with the disappearance of any *background* was a mistake. Einstein himself, in Leiden in 1920, warned that a space devoid of physical properties is unthinkable — no law could be conceived there.¹

The thesis defended here is sober: the background *must* remain, but its nature changes. No longer *mechanical* or *energetic*, still less *metric*; it becomes *rhythmic*. This is the role of the *substrate* as strictly understood in this text: a *condition of cadence and coherence* carried by the Chronon Field $\Phi(x)$, without its own energy and without modifying geometry. The universe does not need a fluid that fills space, but a *periodicity of existence* that ensures its holding.

This reading is compatible with modern physics: the spacetime metric has its own dynamics and gravitational waves — observed in 2015 — supply its tangible signature.² But $\Phi(x)$ is *not* an additional metric modification: it adds no $T_{\mu\nu}$, does not alter light cones, and only reparametrizes *cadences*. Thus “the ether” is resurrected stripped of its illusions: no longer a substance, but a *coherence weave* in which fields tune to each other.

IV — The Chronon Field as rhythmic substrate

$\Phi(x)$ is neither an energy nor a particle: it is a *local cadence of reality*. At every spacetime point, a frequency $[\Phi] = \text{s}^{-1}$ sets the *reconstitution speed* of phenomena. Processes do not “flow” in a universal time: they are *replayed* according to a local *tempo*. Saying that the vacuum “holds” means that $\Phi(x)$ synchronizes physical operations (appearance, propagation, conservation) without altering geometry.

Operationally, $\Phi(x)$ reparametrizes *cadences*, not geodesic trajectories: light cones are unchanged, General Relativity remains intact, and no additional $T_{\mu\nu}$ is introduced. The *substrate* as intended here is therefore strictly *rhythmic*: a condition of *coherence* and *holding*, not a reservoir of energy nor a mechanical medium.

Terminological harmonization. — Throughout the article, *substrate* $\equiv$ *non-energetic rhythmic background* carried by $\Phi(x)$: this avoids any ambiguity with a “medium”, a metric modification, or a vacuum energy density. This reading aligns with the already stated “master laws” (reparametrization of durations, current $J^\mu = \Phi u^\mu$, separation of phase/loss via Γ) by assigning to Φ a purely *cadential* role.

Minimal operational framework (Φ, Γ). We fix the dimensional status and basic relations:

$$\boxed{[\Phi] = \text{s}^{-1}}, \quad \boxed{d\tau = \Phi^{-1}(x) dt}, \quad \boxed{J^\mu = \Phi u^\mu}, \quad \boxed{\partial_\mu J^\mu = \Gamma(x)}.$$

¹A. Einstein, *Ether and the Theory of Relativity*, lecture at the University of Leiden (1920).

²B. P. Abbott *et al.* (LIGO/Virgo), *Phys. Rev. Lett.* **116**, 061102 (2016).

Here u^μ is a reference congruence ($u^\mu u_\mu = -1$), J^μ a phase current, and $\Gamma(x)$ a local rate of coherence loss/source (dimension s^{-1}). On quantum platforms (qubits, ions, cold atoms), coherence dynamics is read through

$$\dot{C} = (i\Phi - \Gamma)C, \quad T_2(x) \simeq \Gamma(x)^{-1},$$

with a phenomenological law

$$\Gamma(x) = \Gamma_{\text{env}}(x) + \xi \Phi(x) + b |\nabla \Phi(x)|,$$

where Γ_{env} collects known environmental losses and ξ, b are dimensionless. Proofs of covariance and dimensional audits are detailed in *Article 2 (Chronon Field and the End of Timeless Physics)* and its formal appendices.

Box — What the *substrate* is not (operational clarification)

- **No new energy.** $\Phi(x)$ carries no proper energy and adds no $T_{\mu\nu}$ term.
- **No modified metric.** Light cones and GR geometry remain unchanged.
- **Strictly operational role.** $\Phi(x)$ reparametrizes *cadences* (holding, synchronization, coherence); it exerts neither force nor pressure.
- **Separation of phase / loss.** Coherence dynamics is read via $\partial_\mu J^\mu = \Gamma(x)$, with Γ encoding losses (environment, weak couplings), distinct from the phase Φ .

Metrological consequence. — If the substrate is rhythmic, its signatures must be sought not as *background energies*, but as *cadence residuals* (clock frequencies, interferometric phases, spectral stabilities) after subtraction of GR, QED, and instrumental models. Dedicated benches (Casimir, Lamb shift, phase noise) will provide targeted *null tests*, with an *expected null signal* at the 10^{-19} level if the “non-energetic” reading is correct.

V — Vacuum experimental benches: null tests at 10^{-19}

If the *substrate* is strictly *rhythmic* and *non-energetic*, it must induce **no** additional force, **no** extra energy shift, **no** effective pressure term beyond established predictions (QED, GR, instrumental systematics). The right observable is therefore a *cadence residual* after subtraction of the relevant models:

$$\left(\frac{\Delta\nu}{\nu} \right)_{\text{res}} \equiv \left(\frac{\Delta\nu}{\nu} \right)_{\text{obs}} - \left(\frac{\Delta\nu}{\nu} \right)_{\text{QED+GR+instr}} = \varepsilon_\Phi \implies \varepsilon_\Phi \stackrel{\text{non-energetic substrate}}{=} 0 \pm 10^{-19}. \quad (1)$$

Goal: a **null test** at the $\sim 10^{-19}$ (fractional) level, with cross-check controls and independent links.

(A) Casimir effect — vacuum pressure (differential) *Principle.* Measure a *double* device (plates or plate–sphere) in parallel, locked to a mechanical/optical reference oscillator; compare with the full QED prediction (geometry, temperature, roughness, patches). *Observable.* Frequency (or phase) ratio tied to the effective deformation/stiffness: $(\Delta\nu/\nu)_{\text{res}}$ as in (1). *Controls.* Sample permutations, polarity inversions, modulation of plate distance d , temperature, Faraday shielding, vibration isolation; *veto* by a twin device at a remote site. *Expected (non-energetic substrate).* **Null signal** at $\lesssim 10^{-19}$ (fractional). Any robust nonzero residual *falsifies* the non-energetic reading or reveals an uncontrolled systematic.

(B) Lamb shift — differential spectroscopy *Principle.* Compare in parallel sensitive transitions (H: 1S–2S, Rydberg; light ions) to an ultra-stable optical reference; apply high-order QED corrections and environmental shifts (Stark/Zeeman/ac Stark, blackbody). *Observable.* Normalized transition frequency: $(\Delta\nu/\nu)_{\text{res}}$ after QED + environment + instrument corrections. *Controls.* Different cells/geometries, ultra-high vacuum, shielded cavities, standard permutations, *double-blind* calibration sequences. *Expected.* **Null at 10^{-19} .** A stable residual correlated with permutations would indicate either an *energetic* background physics or a bias in the experiment.

(C) Phase noise — interferometry and links *Principle.* Compare two oscillators/lasers/clocks via *independent links* (fiber, GNSS, space), and search for an *irreducible common noise* (spectrum $S_\phi(f)$, Allan deviation $\sigma_y(\tau)$) *after* subtracting link models. *Observable.* Cross-correlation of phase/frequency residuals: $\varepsilon_\Phi^{(1)} \leftrightarrow \varepsilon_\Phi^{(2)}$. *Controls.* Redundant and orthogonal links, permutations of optical routes, “witness” mirror, kilometre-to-continent separations, “black” calibration day sequences. *Expected.* **Zero correlation** at $\sim 10^{-19}$ (fractional) over the relevant integrations. Any robust correlation suggests an unidentified common source or a *non-strictly-rhythmic* background physics.

Acceptance / failure criteria (summary)

- **Acceptance (compatibility with rhythmic substrate).** $\varepsilon_\Phi = 0 \pm 10^{-19}$ on (A)(B)(C), stable under permutations, *independent of the link*, persistent after “black day” runs.
- **Failure (falsification of the non-energetic baseline).** Residual $\varepsilon_\Phi \neq 0$ *with a sign*, reproducible, correlated between independent benches or to an extensive quantity (area, distance d) *after* all controls.

In short: if the *substrate* is nothing but a *cadence*, then the vacuum *adds nothing* to measured energies — it adds only a *holding* that these null tests must leave *silent* down to 10^{-19} .

Formal pointer. — Details on the construction of residuals, “anti-gauge” loops, and covariance audits are given in the appendices of *Article 2 (Chronon Field and the End of Timeless Physics)*.

VI — The background as condition, not cause

To rehabilitate the *substrate* is not to restore an archaic causality. The background is not the *cause* of phenomena; it is their *condition of holding*. Without background, no appearance; without rhythm, no duration; without coherence, no world. Classical metaphysics sought a first mover, modern physics seeks dynamical laws; the Chronon Field proposes an *ontology of maintenance*: the real is not *fabricated*, it is *held* — each instant as a beat of local stability, endlessly replayed.

This perspective is compatible with discrete approaches (loops, noncommutative geometries) where continuity is illuminated as a network of events linked by their *internal coherence*. $\Phi(x)$ then models a *synchronization of points of being*: it deflects neither geodesics nor light cones, but sets the *cadence* at which processes reconstitute themselves.

Testable path (coherence $\rightarrow$ cadence). — By separating *phase* and *loss*, coherence dynamics is read via

$$\partial_\mu J^\mu = \Gamma(x), \quad J^\mu = \Phi u^\mu,$$

with a phenomenological law

$$\Gamma(x) = \Gamma_{\text{env}}(x) + \xi \Phi(x) + b |\nabla \Phi(x)|,$$

where ξ, b are dimensionless. Two independent signatures must then *neutralize each other* if the substrate is strictly non-energetic: (i) **Clocks & relativistic geodesy** — frequency residuals $(\Delta\nu/\nu)_{\text{res}}$ *null* at $\sim 10^{-19}$ after subtracting GR+QED+link models; (ii) **Phase interferometry** (Ramsey, Mach–Zehnder, fiber/GNSS/space links) — *no common correlation* between independent devices at $\sim 10^{-19}$. Any *robust* deviation from these null tests would falsify the *rhythmic non-energetic* hypothesis or reveal an uncontrolled systematic.

VII — Conclusion: rehabilitating the fabric

We live in an age of surfaces: speeds, networks, interfaces. But nothing holds without *background*. The real does not float: it rests on an invisible fabric that is neither matter nor mind — a *pulsation*. The Chronon Field, $\Phi(x)$, restores texture to the vacuum: a vacuum full of *rhythm*.

Far from reinstating a metaphysical ether, it proposes a *dynamic, measurable, revisable* background. The vacuum ceases to be silent; it becomes a continuous *beat* upon which the laws improvise their harmony. Just as the tension of a violin’s strings conditions the very possibility of music, the *pulsation of the substrate* conditions the very possibility of the real.

The background has never disappeared: we had only stopped listening to it.

Outlook — This text prepares the following article, *Quantum Rhythm: When Information Breathes in Time*, where the rhythmic fabric is translated into an explicit dynamics of coherence amplitudes $C(t)$, with systematic separation between phase $\Phi(x)$ and loss $\Gamma(x)$, T_2 protocols and entangled pairs.

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